

Personal Finance Analytics Dashboard

Power BI Data Analytics & Business Intelligence Case Study

Synthetic dataset | 37 transactions | 3-month analysis

Portfolio Case Study

01 — Executive Summary

Personal Finance Analytics Dashboard is a Power BI project designed to transform transaction-level financial data into an interactive analytical solution for monitoring personal financial performance and spending behavior.

The project uses a synthetic dataset of 37 financial transactions covering a three-month period in 2026. Data was prepared using Power Query, modeled using a dedicated date dimension, and analyzed through DAX measures.

The final dashboard provides an overview of income, expenses, net cash flow, and savings rate, while enabling detailed analysis of spending categories, monthly trends, and payment methods.

The analysis identified total income of \$10,500, total expenses of \$5,528, and a savings rate of 47.35%. Housing and Food accounted for 78.60% of total expenses.

02 — Business Problem

Personal financial data can become difficult to interpret when transactions are stored as isolated records without an analytical structure. The objective was to transform transaction-level financial data into an interactive Business Intelligence solution that makes financial performance and spending behavior easier to understand.

03 — Objectives & KPIs

- Monitor Total Income, Total Expenses, Net Cash Flow, and Savings Rate.
- Analyze spending distribution and average expense by category.
- Analyze monthly income, expenses, cash flow, and savings behavior.
- Analyze spending by payment method.
- Enable interactive filtering and cross-filtering.

KPI	Definition
Total Income	Total value of transactions classified as Income
Total Expenses	Total value of transactions classified as Expense
Net Cash Flow	Total Income – Total Expenses
Savings Rate	Net Cash Flow ÷ Total Income

04 — Dataset & Data Preparation

The project uses a synthetic personal finance transaction dataset created specifically for this analysis. It contains 37 transaction records and eight analytical fields: Date, Description, Category, Type, Amount, Payment Method, Month, and Notes.

Power Query preparation included source navigation, data-type changes, a custom column, removal of unnecessary columns, column reordering, additional type adjustments, and renaming Correct Date to Date.

05 — Data Model

The solution uses a dimensional data model consisting of tblTransactions and a dedicated DimDate table.

A one-to-many relationship connects DimDate[Date] to tblTransactions[Date]. DimDate contains Date, Month Name, Month Number, Year, and Year Month. Month Name is sorted by Month Number.

DimDate was created dynamically using the minimum and maximum transaction dates.

06 — DAX & Analytical Layer

Eight DAX measures form the analytical layer: Total Income, Total Expenses, Net Cash Flow, Savings Rate, Average Income, Average Expense, Income Transactions, and Expense Transactions.

Key DAX logic

Total Expenses: filters transactions to Type = Expense and sums Amount.

Net Cash Flow: Total Income minus Total Expenses.

Savings Rate: Net Cash Flow divided by Total Income using DIVIDE for safe denominator handling.

The measures dynamically respond to report filter context.

07 — Dashboard Design & User Experience

Financial Overview

Provides KPI cards for Total Income, Total Expenses, Net Cash Flow, and Savings Rate, plus expense distribution, a monthly financial summary, income vs. expenses by month, and a Month slicer.

Spending Analysis

Provides expense distribution by category, average expense by category, monthly expenses, and payment-method analysis.

The two-page structure follows a progressive analytical flow: financial position → spending distribution → monthly behavior → spending behavior → optimization opportunities.

08 — Validation & Quality Assurance

Power BI results were reconciled against the underlying Excel analysis. Key totals, monthly expenses, categories, payment methods, and derived indicators matched.

Functional testing covered single-month filtering, multi-month filtering, category selection, payment-method selection, cross-highlighting, combined filters, and reset behavior.

09 — Key Insights & Business Recommendations

Strong and stable savings performance

Total income was \$10,500, total expenses were \$5,528, and net cash flow was \$4,972, producing a 47.35% savings rate.

Housing is the primary expense driver

Housing represented \$2,850, or 51.56% of total expenses.

Spending is highly concentrated

Housing and Food together represented 78.60% of total expenses.

Monthly expenses remained stable

Expenses declined from \$1,860 in January to \$1,833 in March, a \$27 (1.45%) reduction.

Income remained constant

Monthly income remained at \$3,500 throughout the three-month period.

Savings rate improved slightly

Savings rate increased by 0.77 percentage points from January to March.

Recommendations

- Monitor Housing as the primary cost driver.
- Closely monitor Food spending for optimization opportunities.
- Maintain the current savings discipline.
- Use the current monthly spending level as a baseline for future monitoring.
- Preserve income stability while optimizing the largest expense categories.

10 — Tools & Technologies

- Microsoft Excel — source dataset and preliminary analysis.
- Power Query — data preparation and transformation.
- Microsoft Power BI — modeling, DAX, visualization, and dashboard development.
- DAX — analytical measures.
- GitHub / Markdown — portfolio documentation.

11 — Conclusion & Project Outcome

This project demonstrates the complete process of transforming structured financial transaction data into an interactive Business Intelligence solution.

The final solution combines data preparation, dimensional modeling, DAX calculations, interactive visualization, validation, and business-oriented analysis within a single Power BI project.

The resulting dashboard provides both a high-level view of financial performance and a detailed analysis of spending behavior.

End-to-end workflow: Excel → Power Query → Dimensional Model → DAX → Interactive Dashboard → Validation → Insights → Recommendations